

Directed lines

VECTOR ALGEBRA 339

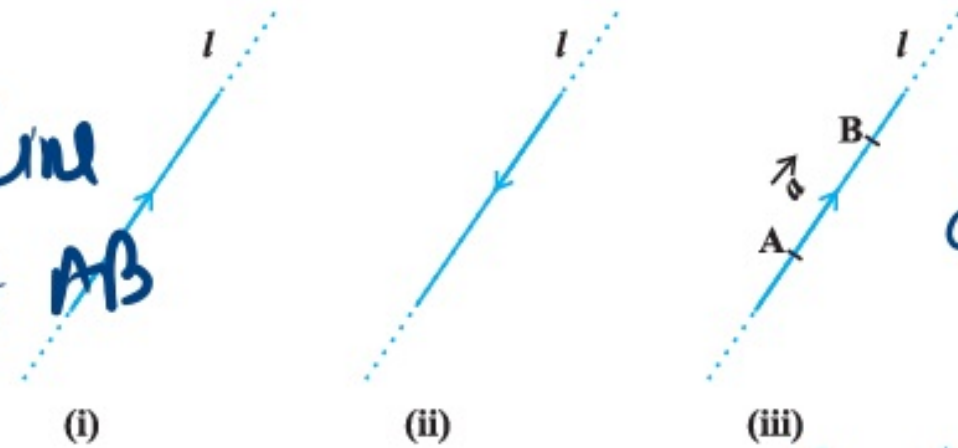


Fig 10.1

Now observe that if we restrict the line l to the line segment AB, then a magnitude is prescribed on the line l with one of the two directions, so that we obtain a *directed line segment* (Fig 10.1(iii)). Thus, a directed line segment has magnitude as well as direction.

Definition 1 A quantity that has magnitude as well as direction is called a vector.

Notice that a directed line segment is a vector (Fig 10.1(iii)), denoted as $\overrightarrow{AB}$ or simply as $\vec{a}$, and read as ‘vector $\overrightarrow{AB}$ ’ or ‘vector $\vec{a}$ ’.

The point A from where the vector $\overrightarrow{AB}$ starts is called its *initial point*, and the point B where it ends is called its *terminal point*. The distance between initial and terminal points of a vector is called the *magnitude* (or length) of the vector, denoted as $|\overrightarrow{AB}|$, or $|\vec{a}|$, or a . The arrow indicates the direction of the vector.

directed line segment = vector

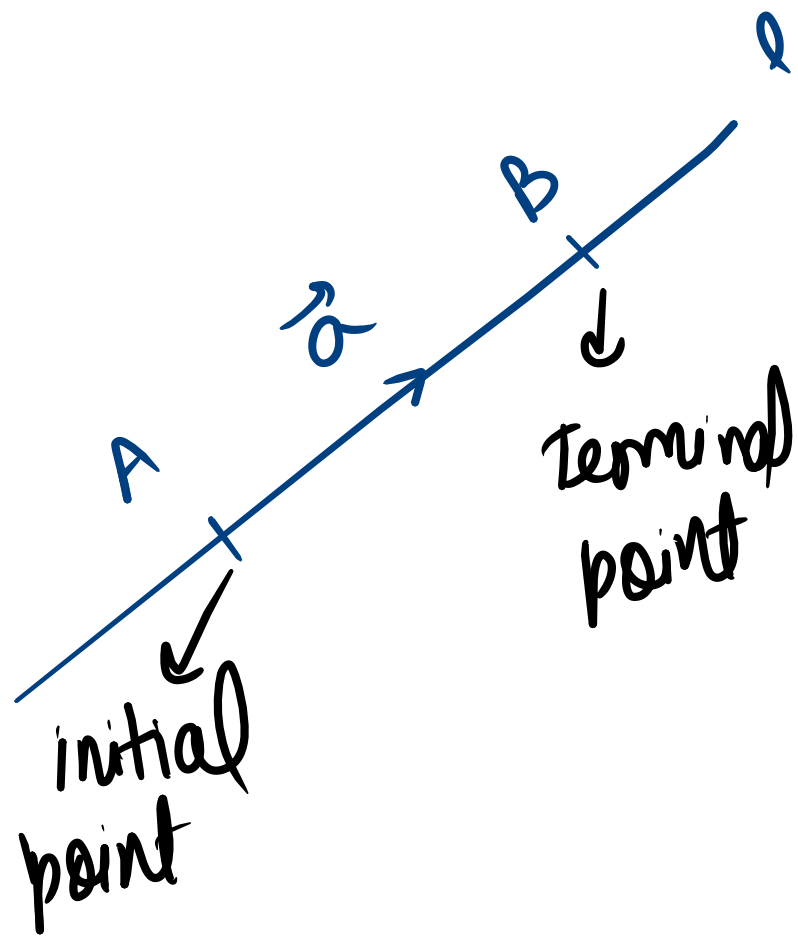
Length of line seg = AB

magnitude of
vector AB

= length of Line Segment AB

Direction of
vector $\vec{AB}$ =

vector $\overrightarrow{AB} = \vec{a}$



$$\text{vector } \vec{AB} = \vec{a}$$

$$\begin{aligned} \text{magnitude of } \vec{AB} &= \text{length of line segment AB} \\ &= |\vec{AB}| = |\vec{a}| = a \end{aligned}$$

$$\text{Direction of } \vec{AB} = \hat{AB}, \hat{a}$$

$$\left[\begin{array}{l} \vec{AB} = |\vec{AB}| \hat{AB} \\ \vec{a} = |\vec{a}| \hat{a} \end{array} \right]$$

Vector is a quantity that has magnitude and direction.
 eg. velocity, displacement, acceleration

Scalar is a quantity that has magn but no direction
 eg weight, height, length

Position vector

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Position of point $P(x, y, z)$ w.r.t origin is called position vector. $\vec{OP} = \vec{r}$

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$$\vec{OP} = |\vec{OP}| \hat{OP}$$

$$\vec{r} = |\vec{r}| \hat{r}$$

$$r = |\vec{OP}| = |\vec{r}| = \sqrt{x^2 + y^2 + z^2}$$

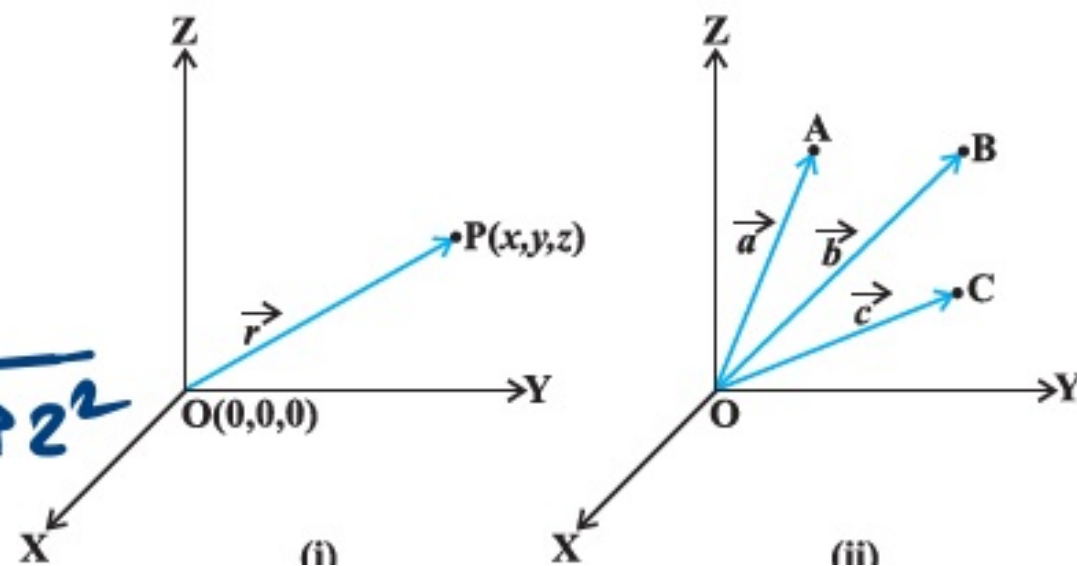


Fig 10.2

Direction Cosines

Consider the position vector $\overline{OP}$ (or $\vec{r}$) of a point $P(x, y, z)$ as in Fig 10.3. The angles α, β, γ made by the vector $\vec{r}$ with the positive directions of x, y and z -axes respectively, are called its *direction angles*. The cosine values of these angles, i.e., $\cos \alpha, \cos \beta$ and $\cos \gamma$ are called *direction cosines* of the vector $\vec{r}$, and usually denoted by l, m and n , respectively.

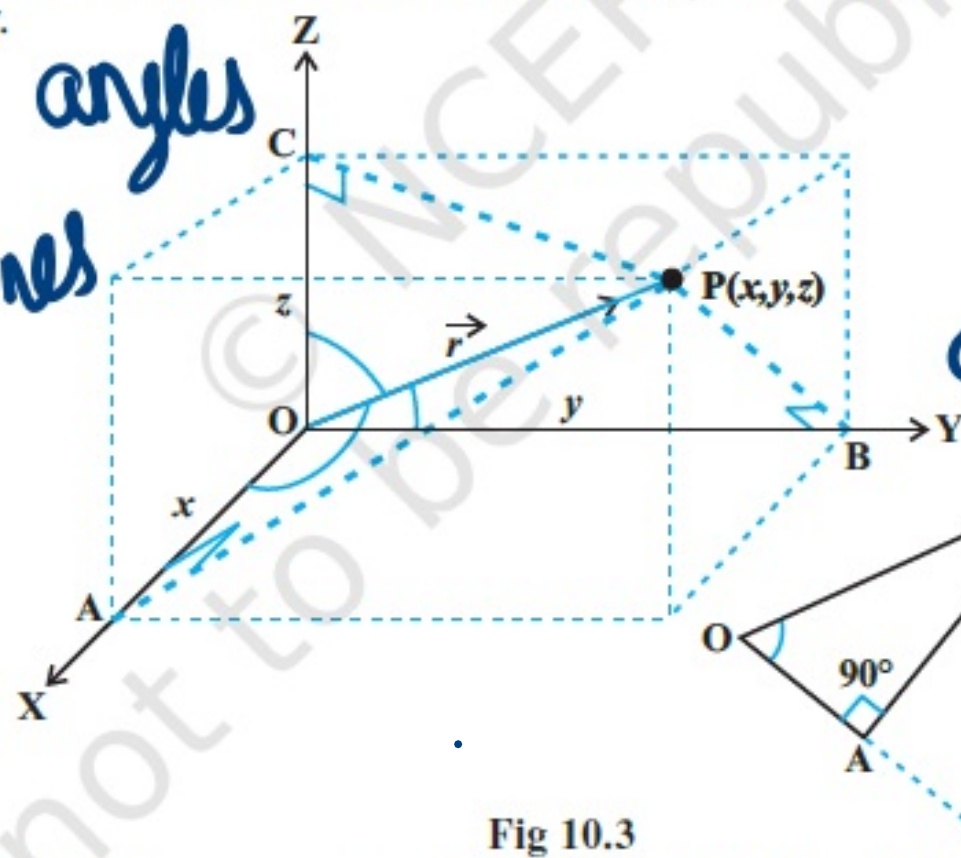


Fig 10.3

From Fig 10.3, one may note that the triangle OAP is right angled, and in it, we have $\cos \alpha = \frac{x}{r}$ (r stands for $|\vec{r}|$). Similarly, from the right angled triangles OBP and OCP, we may write $\cos \beta = \frac{y}{r}$ and $\cos \gamma = \frac{z}{r}$. Thus, the coordinates of the point P may also be expressed as (lr, mr, nr) . The numbers lr, mr and nr , proportional to the direction cosines are called as *direction ratios* of vector $\vec{r}$, and denoted as a, b and c , respectively.

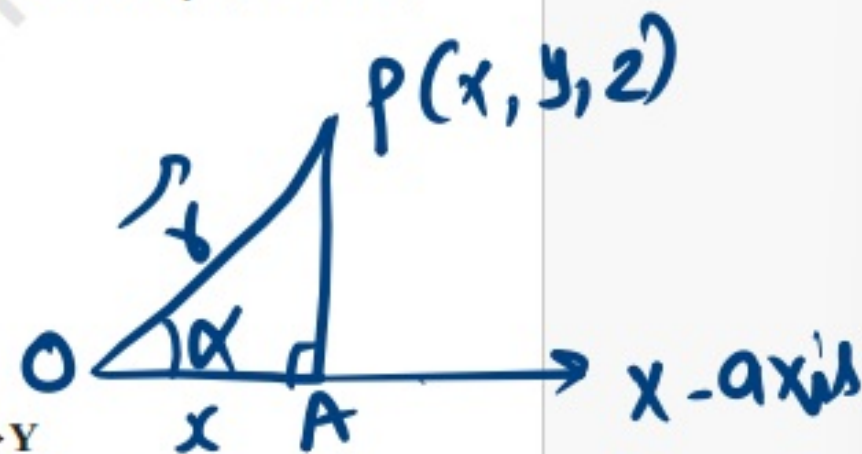
$\alpha, \beta, \gamma \rightarrow$ Direction angles

$\cos \alpha, \cos \beta, \cos \gamma \rightarrow$ Dirⁿ Cosines

$$l = \cos \alpha$$

$$m = \cos \beta$$

$$n = \cos \gamma$$



$$\cos \alpha = \frac{OA}{OP} = \frac{x}{r}$$

$$x = r \cos \alpha$$

$$\begin{aligned} x &= rl \\ y &= rm \\ z &= rn \end{aligned}$$

Note:
1)

$$\boxed{x = rl \quad ; \quad y = rm \quad ; \quad z = rn}$$

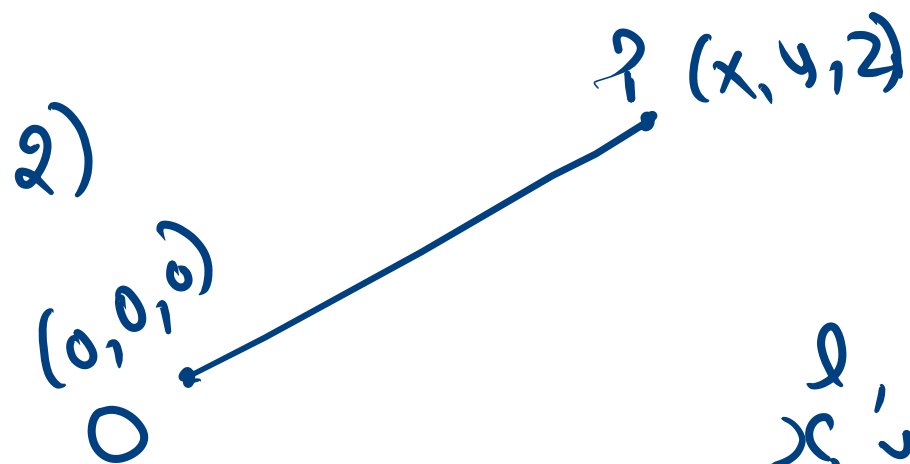
$$\frac{x}{r} = l \quad ; \quad \frac{y}{r} = m \quad ; \quad \frac{z}{r} = n$$

$$l^2 + m^2 + n^2 = \frac{x^2 + y^2 + z^2}{r^2} = \frac{x^2 + y^2 + z^2}{x^2 + y^2 + z^2} = 1$$

$$\boxed{l^2 + m^2 + n^2 = 1}$$

l, m, n are

For position vector

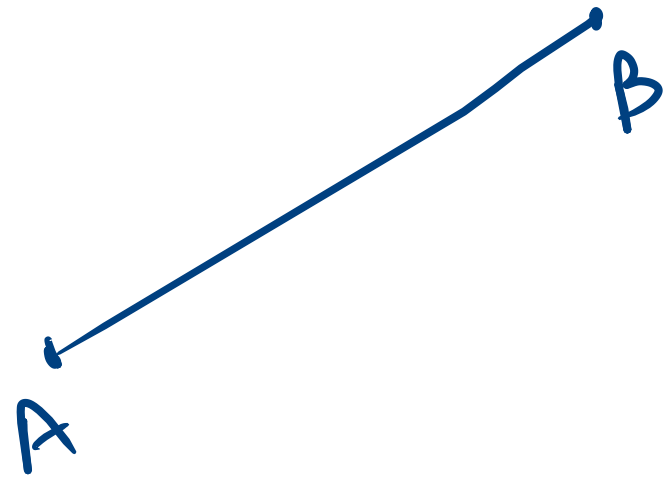


$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n} = r$$

$l, m, n \rightarrow$ D.C's
 $x, y, z \rightarrow$ D.R's

D.R's and D.C's are proportional to each other.

3)



for any vector $\vec{AB}$
D.R's and D.C's of $\vec{AB}$ are
proportional to each other

D.R's of a general vector $\vec{AB}$ are denoted as a, b, c

D.C's of a general vector $\vec{AB}$ are denoted as l, m, n

$$l = \cos \alpha, \quad m = \cos \beta, \quad n = \cos \gamma$$

$\alpha, \beta, \gamma \rightarrow$ Direction angles

angle $\swarrow$ made by vector with x-axis

Note One may note that $l^2 + m^2 + n^2 = 1$ but $a^2 + b^2 + c^2 \neq 1$, in general.

10.3 Types of Vectors

Zero Vector A vector whose initial and terminal points coincide, is called a zero vector (or null vector), and denoted as $\vec{0}$. Zero vector can not be assigned a definite direction as it has zero magnitude. Or, alternatively otherwise, it may be regarded as having any direction. The vectors $\overline{AA}$, $\overline{BB}$ represent the zero vector,

Unit Vector A vector whose magnitude is unity (i.e., 1 unit) is called a unit vector. The unit vector in the direction of a given vector $\vec{a}$ is denoted by $\hat{a}$.

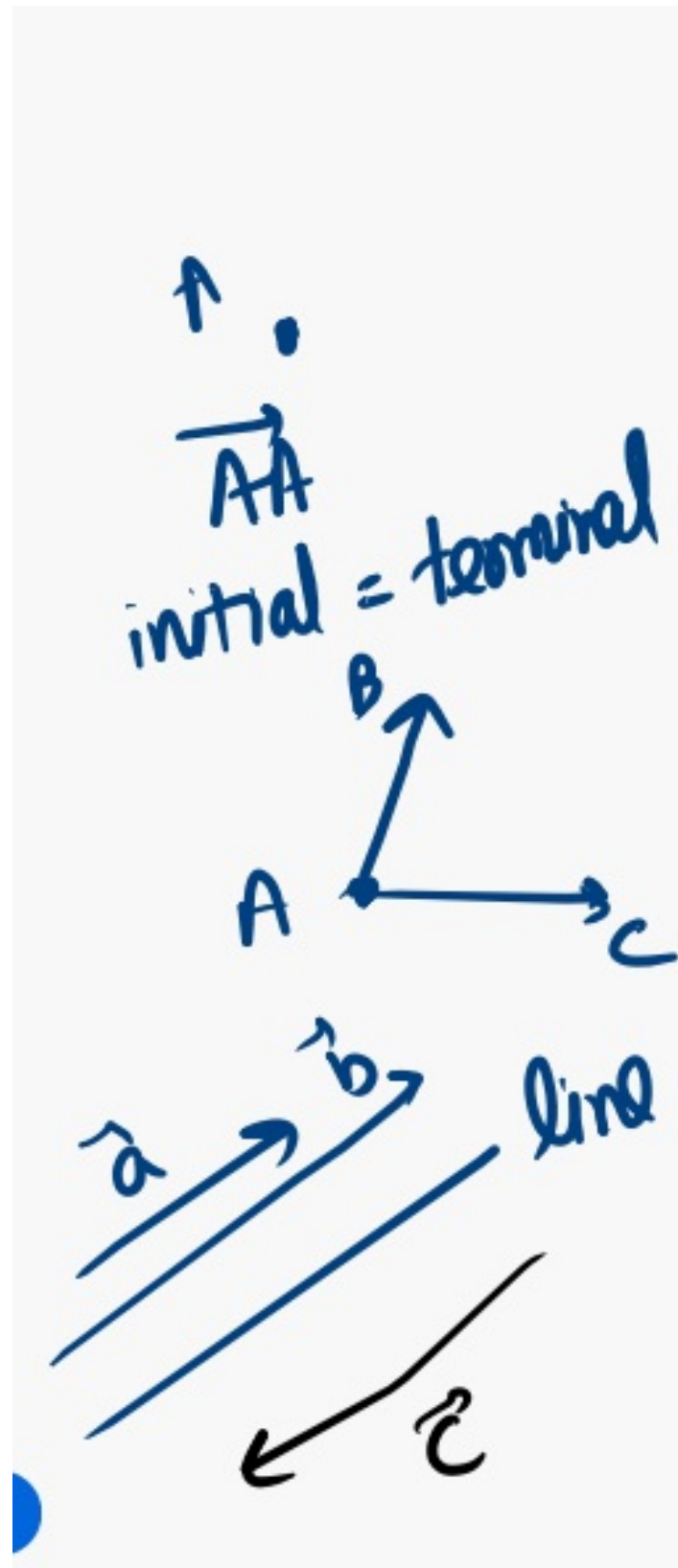
Coinitial Vectors Two or more vectors having the same initial point are called coinital vectors.

Collinear Vectors Two or more vectors are said to be collinear if they are parallel to the same line, irrespective of their magnitudes and directions.

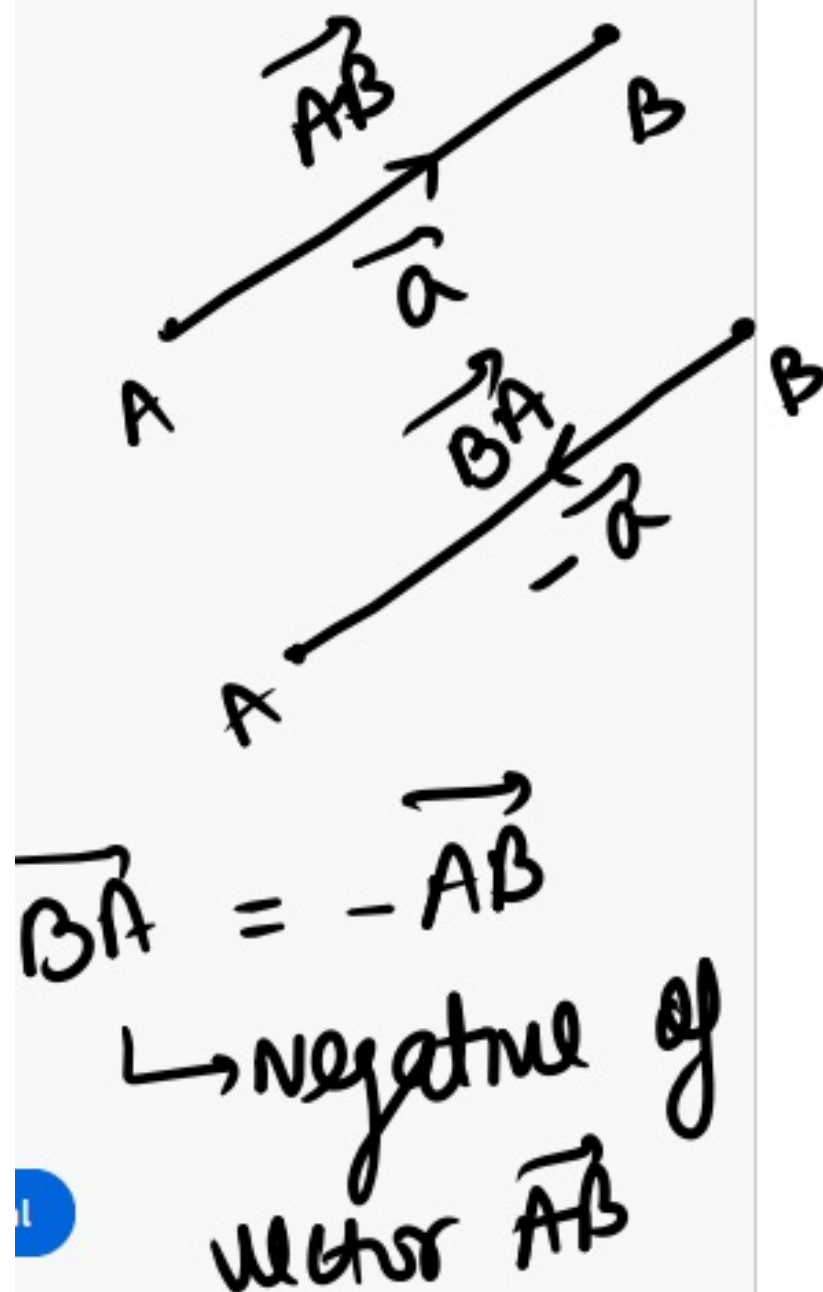
Equal Vectors Two vectors $\vec{a}$ and $\vec{b}$ are said to be equal, if they have the same magnitude and direction regardless of the positions of their initial points, and written as $\vec{a} = \vec{b}$.

Negative of a Vector A vector whose magnitude is the same as that of a given vector (say, $\overline{AB}$), but direction is opposite to that of it, is called *negative* of the given vector. For example, vector $\overline{BA}$ is negative of the vector $\overline{AB}$, and written as $\overline{BA} = -\overline{AB}$.

Remark The vectors defined above are such that any of them may be subject to its parallel displacement without changing its magnitude and direction. Such vectors are called *free vectors*. Throughout this chapter, we will be dealing with free vectors only.



$\vec{a} \rightarrow$ unit vector
 $\Rightarrow |\vec{a}| = 1$



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Triangle law of vector addition

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$$\vec{AB} + \vec{BC} + \vec{CA} = \vec{0} \rightarrow \text{zero vector}$$

$$\vec{AB} + \vec{BC} = -\vec{CA}$$

$$\vec{AB} + \vec{BC} = \vec{AC}$$

$$\vec{a} + \vec{b} = \vec{AC}$$

10.4 Addition of Vectors

A vector $\vec{AB}$ simply means the displacement from a point A to the point B. Now consider a situation that a girl moves from A to B and then from B to C (Fig 10.7). The net displacement made by the girl from point A to the point C, is given by the vector $\vec{AC}$ and expressed as

$$\vec{AC} = \vec{AB} + \vec{BC}$$

This is known as the triangle law of vector addition.

In general, if we have two vectors $\vec{a}$ and $\vec{b}$ (Fig 10.8 (i)), then to add them, they are positioned so that the initial point of one coincides with the terminal point of the other (Fig 10.8(ii)).

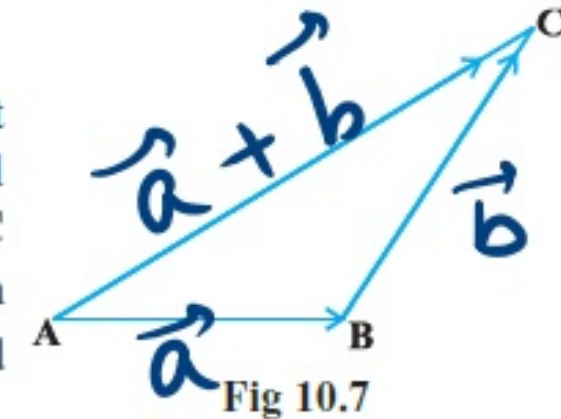
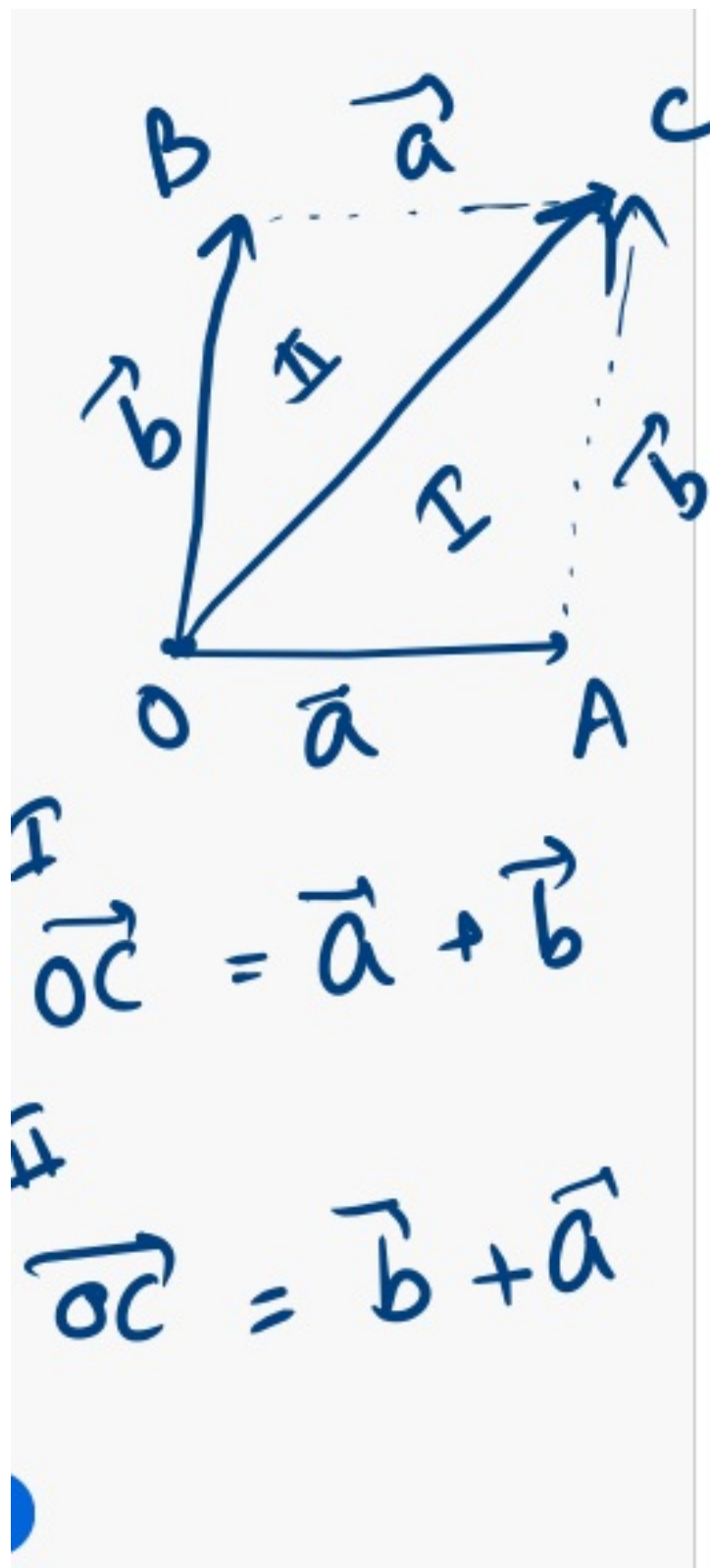


Fig 10.7

Parallelogram



direction perpendicular to the flow of the river then, it is acted upon by two velocity vectors—one is the velocity imparted to the boat by its engine and other one is the velocity of the flow of river water. Under the simultaneous influence of these two velocities, the boat in actual starts travelling with a different velocity. To have a precise idea about the effective speed and direction (i.e., the resultant velocity) of the boat, we have the following law of vector addition.

If we have two vectors $\vec{a}$ and $\vec{b}$ represented by the two adjacent sides of a parallelogram in magnitude and direction (Fig 10.9), then their sum $\vec{a} + \vec{b}$ is represented in magnitude and direction by the diagonal of the parallelogram through their common point. This is known as the parallelogram law of vector addition.

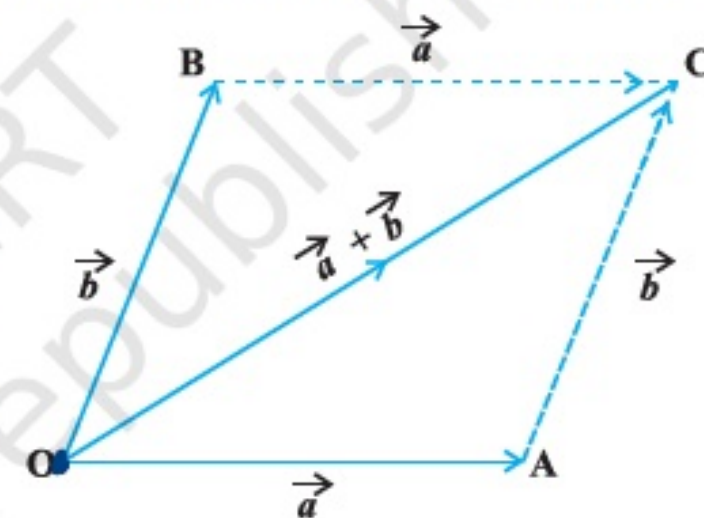


Fig 10.9

Note From Fig 10.9, using the triangle law, one may note that

$$\vec{OA} + \vec{AC} = \vec{OC}$$

or

$$\vec{OA} + \vec{OB} = \vec{OC} \quad (\text{since } \vec{AC} = \vec{OB})$$

which is parallelogram law. Thus, we may say that the two laws of vector addition are equivalent to each other.

Properties of vector addition

Property 1 For any two vectors $\vec{a}$ and $\vec{b}$,

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

(Commutative property)

Properties

i) Commutative Property

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

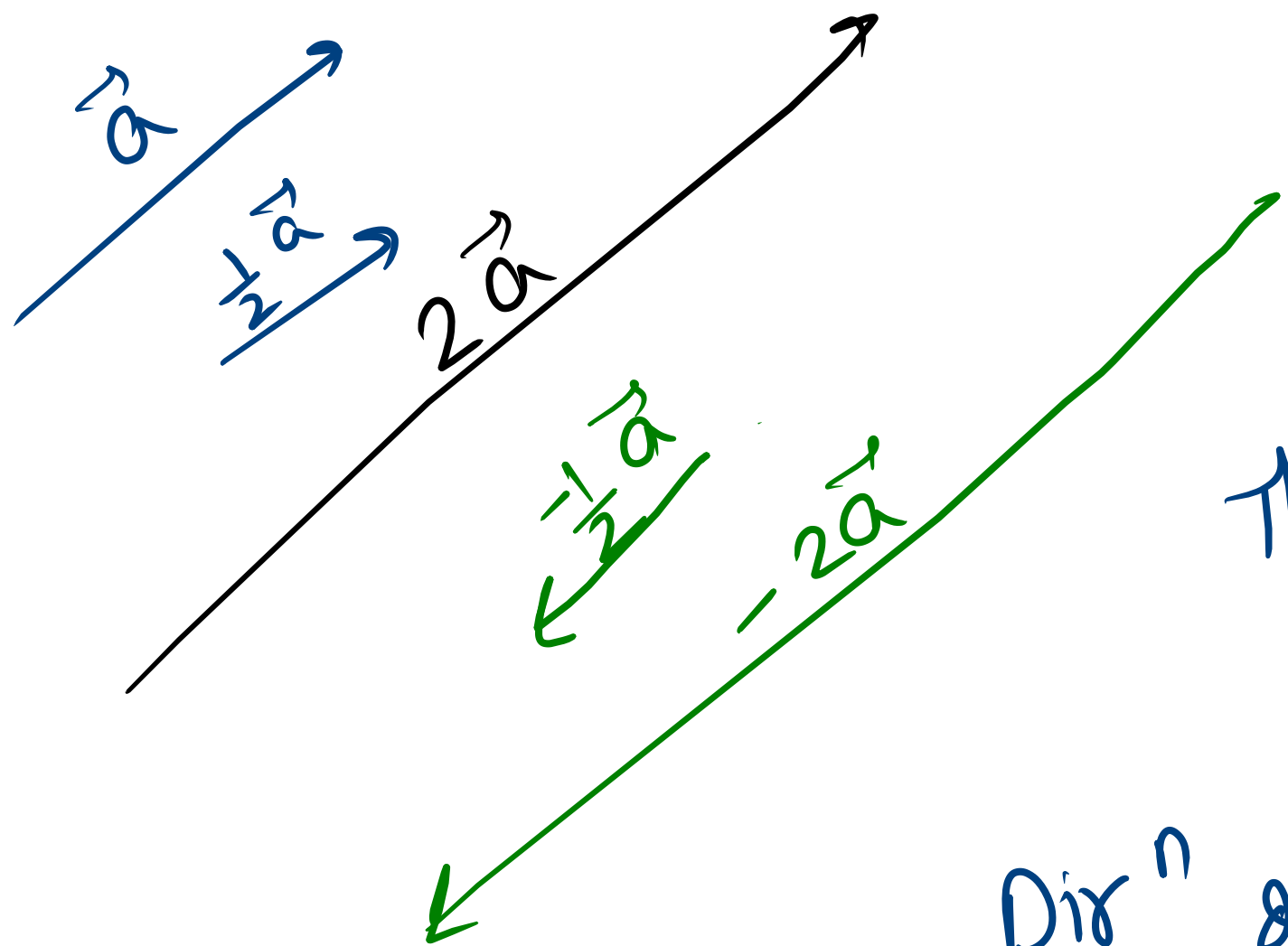
ii) Associative

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$$

iii) $\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$

iv)

iv) Multiplication of a vector by a scalar



$$\lambda \vec{a}$$

$\lambda \rightarrow \text{scalar}$

Then magⁿ of new vector $\lambda \vec{a}$
 $= |\lambda \vec{a}| = |\lambda| |\vec{a}|$

Dirⁿ of new vector $\lambda \vec{a}$ $\begin{cases} \rightarrow \text{same as } \vec{a} & \text{if } \lambda > 0 \\ \rightarrow \text{opp to } \vec{a} & \text{if } \lambda < 0 \end{cases}$

v) unit vector in dirⁿ of $\vec{a}$

$$\vec{a} = |\vec{a}| \hat{a}$$

$$1 \cdot \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \text{unit vector in the dir}^n \text{ of } \vec{a}$$

find unit vector in dirⁿ of $\vec{AC}$

→ you will find $\hat{AC} = \frac{\vec{AC}}{|\vec{AC}|}$

$\hat{i}$ → unit vector
along x-axis

$\hat{j}$ → unit vector
along y-axis

$\hat{k}$ → unit vector
along z-axis

$$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{OP_1} = x\hat{i} + y\hat{j}$$

In $\triangle OP_1P$,

$$\vec{P_1P} = \vec{OR} = z\hat{k}$$

10.5.1 Components of a vector

Let us take the points $A(1, 0, 0)$, $B(0, 1, 0)$ and $C(0, 0, 1)$ on the x -axis, y -axis and z -axis, respectively. Then, clearly

$$|\vec{OA}| = 1, |\vec{OB}| = 1 \text{ and } |\vec{OC}| = 1$$

The vectors $\vec{OA}$, $\vec{OB}$ and $\vec{OC}$, each having magnitude 1, are called *unit vectors along the axes* OX , OY and OZ , respectively, and denoted by $\hat{i}$, $\hat{j}$ and $\hat{k}$, respectively (Fig 10.13).

Now, consider the position vector $\vec{OP}$ of a point $P(x, y, z)$ as in Fig 10.14. Let P_1 be the foot of the perpendicular from P on the plane XOY .

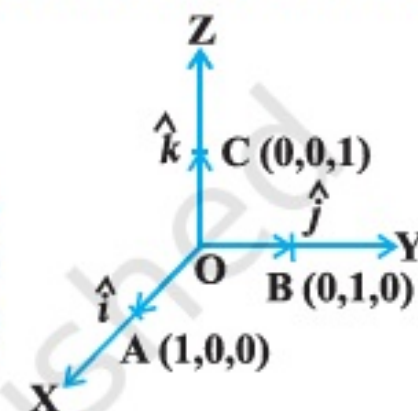


Fig 10.13

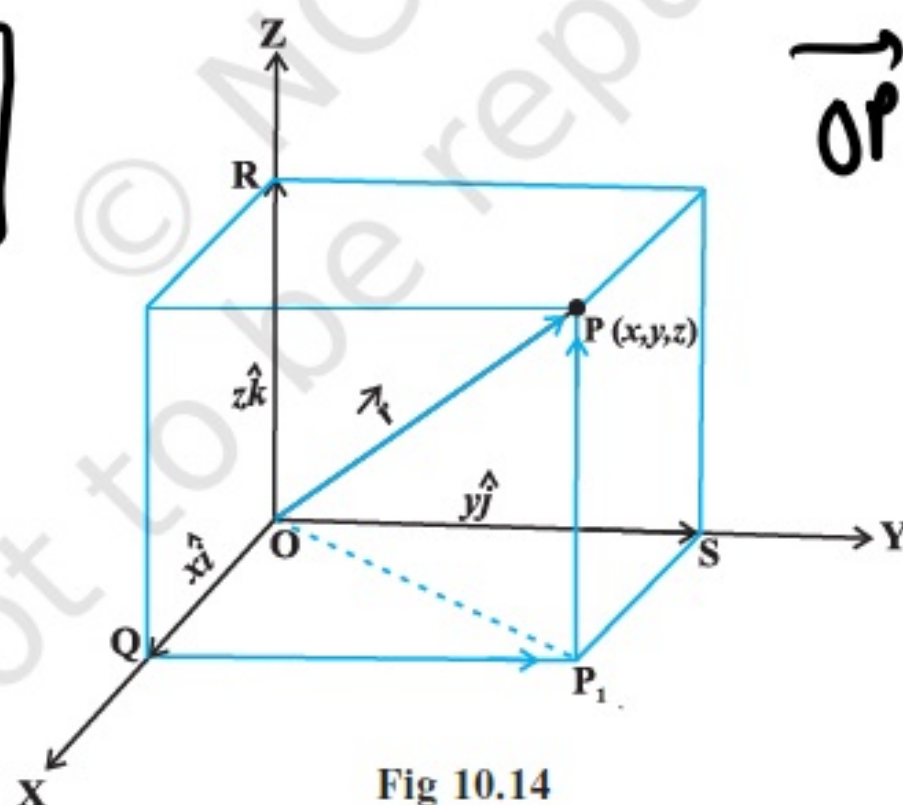


Fig 10.14

We, thus, see that P_1P is parallel to z -axis. As $\hat{i}$, $\hat{j}$ and $\hat{k}$ are the unit vectors along the x , y and z -axes, respectively, and by the definition of the coordinates of P , we have

$$\begin{aligned}\vec{OP} &= \vec{OP_1} + \vec{P_1P} \\ &= x\hat{i} + y\hat{j} + z\hat{k}\end{aligned}$$

$$|\vec{OI}| = |\vec{r}| = r = \sqrt{x^2 + y^2 + z^2} = \text{mag}^n \text{ of vector } \vec{OP}$$

eg ✓ $\vec{OP} = 2\hat{i} - 3\hat{j} + 4\hat{k}$

$$|\vec{OP}| = \sqrt{2^2 + (-3)^2 + 4^2} = \sqrt{29}$$

Coordinates of point P $\equiv (2, -3, 4)$

$$\text{unit vector in dir}^n \text{ of } \vec{OP} = \hat{OP} = \frac{\vec{OP}}{|\vec{OP}|} = \frac{2\hat{i} - 3\hat{j} + 4\hat{k}}{\sqrt{29}}$$

Properties on components of a vector

$$\overrightarrow{OA} = \vec{a} = a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k}$$

$$\overrightarrow{OB} = \vec{b} = a_2 \hat{i} + b_2 \hat{j} + c_2 \hat{k}$$

i) $\vec{a} + \vec{b} = (a_1 + a_2) \hat{i} + (b_1 + b_2) \hat{j} + (c_1 + c_2) \hat{k}$

ii) $\vec{a} - \vec{b} = (a_1 - a_2) \hat{i} + (b_1 - b_2) \hat{j} + (c_1 - c_2) \hat{k}$

iii) $2\vec{a} = 2a_1 \hat{i} + 2b_1 \hat{j} + 2c_1 \hat{k}$

iv) $\vec{a}$ & $\vec{b}$ are equal

$$a_1 = a_2 ; b_1 = b_2 ; c_1 = c_2$$

$$v) \quad (p + q) \vec{a} = p \vec{a} + q \vec{a}$$

$$vi) \quad (p q) \vec{a} = p (q \vec{a}) = q (p \vec{a})$$

$$\underline{\underline{vii)}} \quad p (\vec{a} + \vec{b}) = p \vec{a} + p \vec{b}$$

Example 4 Find the values of x , y and z so that the vectors $\vec{a} = x\hat{i} + 2\hat{j} + z\hat{k}$ and $\vec{b} = 2\hat{i} + y\hat{j} + \hat{k}$ are equal.

sol

$$\vec{a} = \vec{b}$$
$$\underline{x\hat{i}} + \underline{2\hat{j}} + z\hat{k} = \underline{2\hat{i}} + y\hat{j} + \hat{k}$$

$$x = 2$$

$$y = 2$$

$$z = 1$$

Example 5 Let $\vec{a} = \hat{i} + 2\hat{j}$ and $\vec{b} = 2\hat{i} + \hat{j}$. Is $|\vec{a}| = |\vec{b}|$? Are the vectors $\vec{a}$ and $\vec{b}$ equal?

sol

$$|\vec{a}| = \sqrt{1^2 + 2^2 + 0^2} = \sqrt{5}$$

$$|\vec{b}| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$|\vec{a}| = |\vec{b}| \quad \underline{\text{Yes}}$$

No, vectors are not equal, because their components are not same.

Example 6 Find unit vector in the direction of vector $\vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}$

sol unit vector in dirⁿ of $\vec{a} = \hat{a} = \frac{\vec{a}}{|\vec{a}|}$

$$|\vec{a}| = \sqrt{2^2 + 3^2 + 1^2} = \sqrt{14}$$

$$\hat{a} = \frac{2\hat{i} + 3\hat{j} + \hat{k}}{\sqrt{14}} = \frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}$$

Example 7 Find a vector in the direction of vector $\vec{a} = \hat{i} - 2\hat{j}$ that has magnitude 7 units.

sol

let the vector be $\vec{b}$ $\begin{cases} |\vec{b}| = 7 \\ \hat{b} = ? \end{cases}$

$\hat{b}$ = direction of vector $\vec{a}$

$$\hat{b} = \hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} - 2\hat{j}}{\sqrt{1^2 + (-2)^2}} = \frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j}$$

$$\vec{b} = |\vec{b}| \hat{b} = 7 \left(\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{j} \right) = \frac{7}{\sqrt{5}}\hat{i} - \frac{14}{\sqrt{5}}\hat{j}$$

Example 8 Find the unit vector in the direction of the sum of the vectors, $\vec{a} = 2\hat{i} + 2\hat{j} - 5\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} + 3\hat{k}$.

$$\text{(Ans)} \quad \vec{c} = \vec{a} + \vec{b} = 4\hat{i} + 3\hat{j} - 2\hat{k}$$

$$\hat{c} = \frac{\vec{c}}{|\vec{c}|}$$

Example 9 Write the direction ratio's of the vector $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$ and hence calculate its direction cosines.

$$\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$$

$$\text{D.R's of } \vec{a} = (1, 1, -2)$$

$$|\vec{a}| = \sqrt{1^2 + 1^2 + (-2)^2} = \sqrt{6}$$

$$\text{D.C's of } \vec{a} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}} \right)$$

Homework

Exercise - 10.1

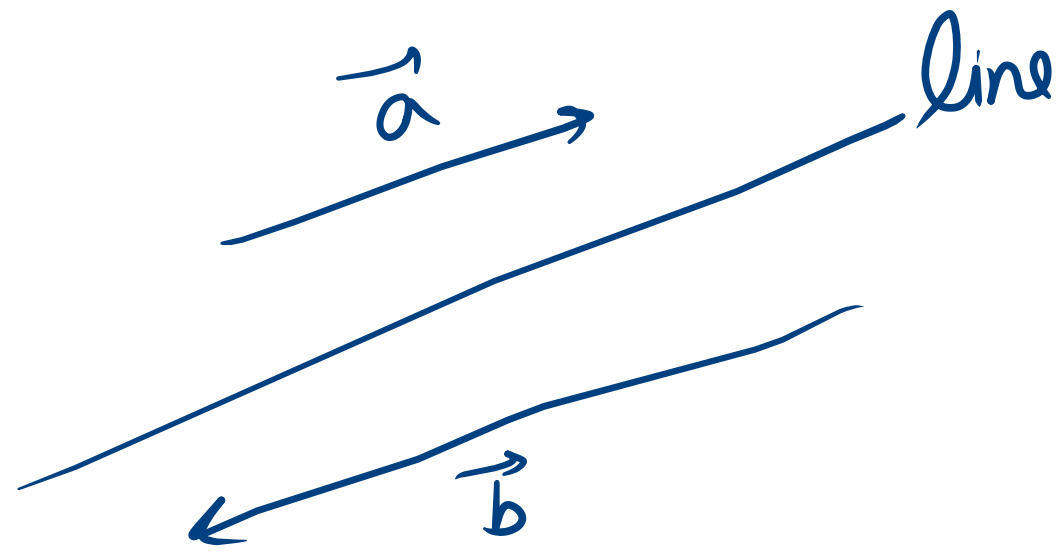
Exercise - 10.2 $\rightarrow$ 01, 04, 06, 07, 09, 010, 012, 018

Example - 6, 7, 8, 9

+ Previous class left over questions.

Property of components of vector

(viii) $\vec{a} = a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k}$
 $\vec{b} = a_2 \hat{i} + b_2 \hat{j} + c_2 \hat{k}$



if $\vec{a}$ & $\vec{b}$ are collinear

$$\hat{a} = K_2 \hat{b} \quad K_2 = \pm 1$$

$$|\vec{a}| = K_1 |\vec{b}|$$

$$\vec{a} = |\vec{a}| \hat{a} = K_1 |\vec{b}| K_2 \hat{b}$$

$$\vec{a} = K_1 K_2 |\vec{b}| \hat{b}$$

then

$$\boxed{\vec{a} = \lambda \vec{b}} \quad \lambda \rightarrow \text{scalar, constant}$$

$$\vec{a} = \lambda \vec{b}$$

$$a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k} = \lambda a_2 \hat{i} + \lambda b_2 \hat{j} + \lambda c_2 \hat{k}$$

$$\lambda a_2 = a_1 ; \quad \lambda b_2 = b_1 ; \quad \lambda c_2 = c_1$$

$$\lambda = \frac{a_1}{a_2} ; \quad \lambda = \frac{b_1}{b_2} ; \quad \lambda = \frac{c_1}{c_2}$$

$$\boxed{\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}}$$

$$\boxed{\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}}$$

→ If two vectors are collinear then their components/D.Rs/D.Cs will be proportional.

Q11
D.R of $\vec{a} = (2, -3, 4)$
D.R of $\vec{b} = (-4, 6, -8)$

$$-\frac{2}{4} = -\frac{1}{2}$$

$$\frac{-3}{6} = -\frac{1}{2}$$

$$\frac{4}{-8} = -\frac{1}{2}$$

their D.R are proportional
 $\Rightarrow$ they are collinear

5. Find the scalar and vector components of the vector with initial point (2, 1) and terminal point (-5, 7).
6. Find the sum of the vectors $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{c} = \hat{i} - 6\hat{j} - 7\hat{k}$.
7. Find the unit vector in the direction of the vector $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$.
8. Find the unit vector in the direction of vector $\overline{PQ}$, where P and Q are the points (1, 2, 3) and (4, 5, 6), respectively.
9. For given vectors, $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} - \hat{k}$, find the unit vector in the direction of the vector $\vec{a} + \vec{b}$.
10. Find a vector in the direction of vector $5\hat{i} - \hat{j} + 2\hat{k}$ which has magnitude 8 units.
11. Show that the vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $-4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear.
12. Find the direction cosines of the vector $\hat{i} + 2\hat{j} + 3\hat{k}$.
13. Find the direction cosines of the vector joining the points A (1, 2, -3) and B (-1, -2, 1), directed from A to B.
14. Show that the vector $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the axes OX, OY and OZ.
15. Find the position vector of a point R which divides the line joining two points P and Q whose position vectors are $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively, in the ratio 2 : 1
 - (i) internally
 - (ii) externally

8. Find the unit vector in the direction of vector $\overrightarrow{PQ}$, where P and Q are the points (1, 2, 3) and (4, 5, 6), respectively.

$$\begin{aligned}\overrightarrow{PQ} &= (4-1)\hat{i} + (5-2)\hat{j} + (6-3)\hat{k} \\ &= 3\hat{i} + 3\hat{j} + 3\hat{k}\end{aligned}$$

$$|\overrightarrow{PQ}| = \sqrt{3^2 + 3^2 + 3^2} = 3\sqrt{3}$$

$$\begin{aligned}\hat{PQ} &= \frac{\overrightarrow{PQ}}{|\overrightarrow{PQ}|} = \frac{3}{3\sqrt{3}}\hat{i} + \frac{3}{3\sqrt{3}}\hat{j} + \frac{3}{3\sqrt{3}}\hat{k} \\ &= \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}\end{aligned}$$



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Q5
 $\vec{PO} = -7\hat{i} + 6\hat{j} + 0\hat{k}$
 vector comp = $-7\hat{i}, 6\hat{j}, 0\hat{k}$
 scalar comp = $-7, 6, 0$

1. Compute the magnitude of the following vectors:

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}; \quad \vec{b} = 2\hat{i} - 7\hat{j} - 3\hat{k}; \quad \vec{c} = \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} - \frac{1}{\sqrt{3}}\hat{k}$$

2. Write two different vectors having same magnitude.

3. Write two different vectors having same direction.

4. Find the values of x and y so that the vectors $2\hat{i} + 3\hat{j}$ and $x\hat{i} + y\hat{j}$ are equal.

5. Find the scalar and vector components of the vector with initial point (2, 1) and terminal point (-5, 7).

6. Find the sum of the vectors $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{c} = \hat{i} - 6\hat{j} - 7\hat{k}$.

7. Find the unit vector in the direction of the vector $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$.

8. Find the unit vector in the direction of vector $\vec{PQ}$, where P and Q are the points (1, 2, 3) and (4, 5, 6), respectively.

9. For given vectors, $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} - \hat{k}$, find the unit vector in the direction of the vector $\vec{a} + \vec{b}$.

10. Find a vector in the direction of vector $5\hat{i} - \hat{j} + 2\hat{k}$ which has magnitude 8 units.

11. Show that the vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $-4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear.

12. Find the direction cosines of the vector $\hat{i} + 2\hat{j} + 3\hat{k}$.

13. Find the direction cosines of the vector joining the points A (1, 2, -3) and B (-1, -2, 1), directed from A to B.

14. Show that the vector $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the axes OX, OY and OZ.

15. Find the position vector of a point R which divides the line joining two points P and Q whose position vectors are $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively, in the ratio 2 : 1

Free trial

Q13

$$\vec{AB} = -2\hat{i} + 4\hat{j} + 4\hat{k}$$

$$D.R \equiv (-2, -4, +4)$$

$$|\vec{AB}| = 9$$

$$D.Cs = \left(\frac{-2}{9}, \frac{-4}{9}, \frac{4}{9} \right)$$

1. Compute the magnitude of the following vectors:

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}; \quad \vec{b} = 2\hat{i} - 7\hat{j} - 3\hat{k}; \quad \vec{c} = \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} - \frac{1}{\sqrt{3}}\hat{k}$$

2. Write two different vectors having same magnitude.

3. Write two different vectors having same direction.

4. Find the values of x and y so that the vectors $2\hat{i} + 3\hat{j}$ and $x\hat{i} + y\hat{j}$ are equal.

5. Find the scalar and vector components of the vector with initial point $(2, 1)$ and terminal point $(-5, 7)$.

6. Find the sum of the vectors $\vec{a} = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} + 5\hat{k}$ and $\vec{c} = \hat{i} - 6\hat{j} - 7\hat{k}$.

7. Find the unit vector in the direction of the vector $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$.

8. Find the unit vector in the direction of vector $\vec{PQ}$, where P and Q are the points $(1, 2, 3)$ and $(4, 5, 6)$, respectively.

9. For given vectors, $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} - \hat{k}$, find the unit vector in the direction of the vector $\vec{a} + \vec{b}$.

10. Find a vector in the direction of vector $5\hat{i} - \hat{j} + 2\hat{k}$ which has magnitude 8 units.

11. Show that the vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $-4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear.

12. Find the direction cosines of the vector $\hat{i} + 2\hat{j} + 3\hat{k}$.

13. Find the direction cosines of the vector joining the points $A(1, 2, -3)$ and $B(-1, -2, 1)$, directed from A to B .

14. Show that the vector $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the axes OX , OY and OZ .

15. Find the position vector of a point R which divides the line joining two points P and Q whose position vectors are $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively, in the ratio $2 : 1$

Q14

Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

equally inclined to

all axis means

Direction angles

$\alpha, \beta, \gamma \rightarrow$ equal

D.Cs = $l, m, n \rightarrow$ equal

9. For given vectors, $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} - \hat{k}$, find the unit vector in the direction of the vector $\vec{a} + \vec{b}$.
10. Find a vector in the direction of vector $5\hat{i} - \hat{j} + 2\hat{k}$ which has magnitude 8 units.
11. Show that the vectors $2\hat{i} - 3\hat{j} + 4\hat{k}$ and $-4\hat{i} + 6\hat{j} - 8\hat{k}$ are collinear.
12. Find the direction cosines of the vector $\hat{i} + 2\hat{j} + 3\hat{k}$.
13. Find the direction cosines of the vector joining the points A (1, 2, -3) and B (-1, -2, 1), directed from A to B.
14. Show that the vector $\hat{i} + \hat{j} + \hat{k}$ is equally inclined to the axes OX, OY and OZ.
15. Find the position vector of a point R which divides the line joining two points P and Q whose position vectors are $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively, in the ratio 2 : 1
 - (i) internally
 - (ii) externally

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16. Find the position vector of the mid point of the vector joining the points P(2, 3, 4) and Q(4, 1, -2).
17. Show that the points A, B and C with position vectors, $\vec{a} = 3\hat{i} - 4\hat{j} - 4\hat{k}$,

$$|\vec{a}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\cos \alpha = \frac{1}{\sqrt{3}}$$

$$\cos \beta = \frac{1}{\sqrt{3}}$$

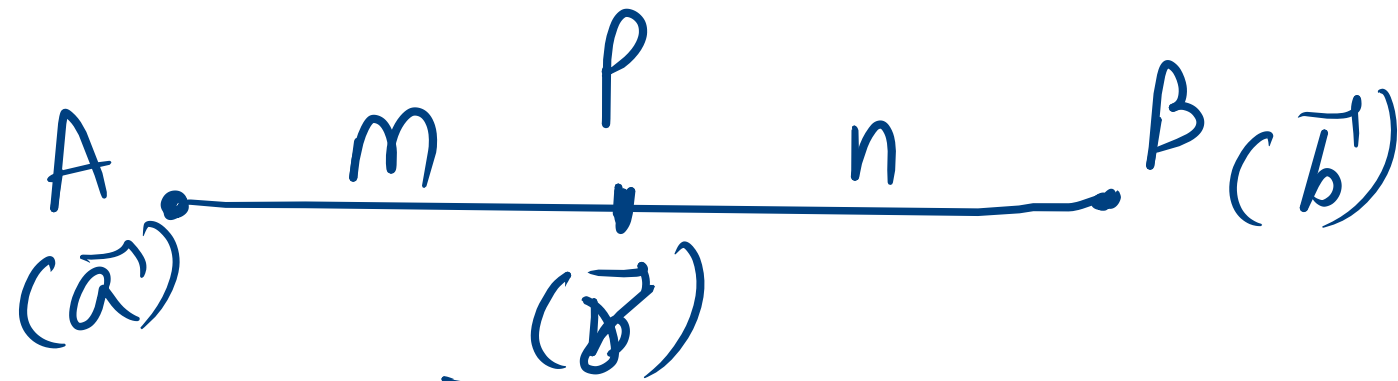
$$\cos \gamma = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \alpha = \beta = \gamma$$

Hence proved

Section Formula

(I) Internal



$$\frac{AP}{BP} = \frac{m}{n}$$

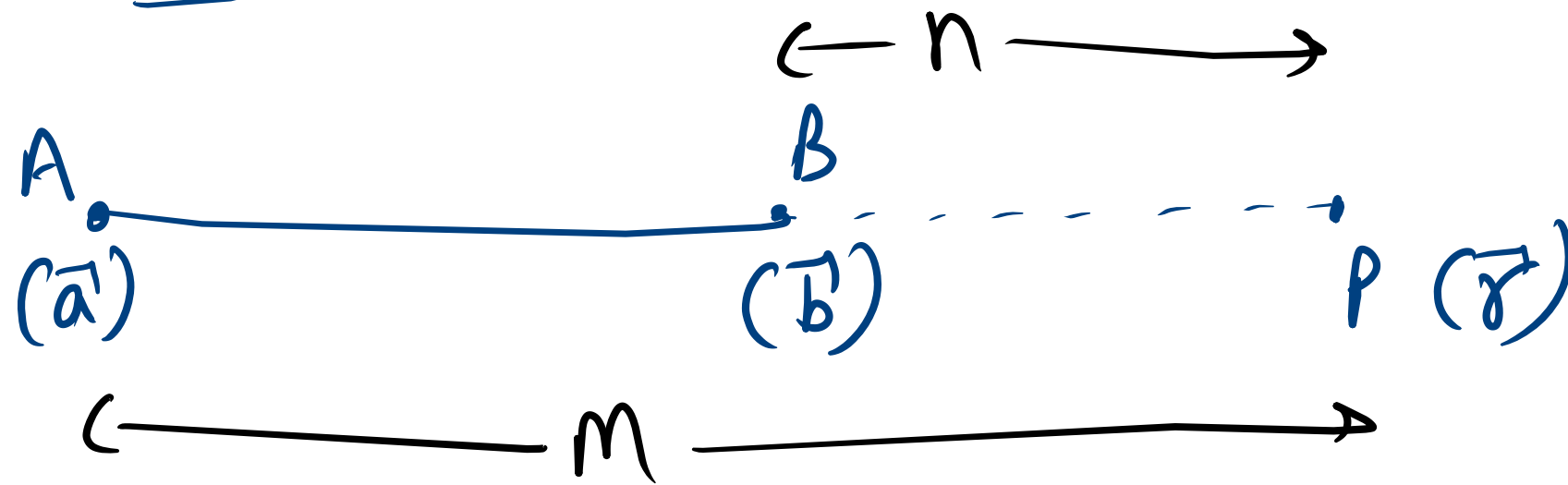
$$\left[\vec{r} = \frac{m\vec{b} + n\vec{a}}{m + n} \right] \checkmark$$

P $\rightarrow$ mid-point

$$\frac{AP}{BP} = \frac{1}{1} = \frac{m}{n}$$

$$\left[\vec{r} = \frac{\vec{b} + \vec{a}}{2} \right] \text{ mid-pt. formula}$$

(II) External section formula



$$\frac{AP}{BP} = \frac{m}{n}$$

$$[\vec{x}] = \frac{m\vec{b} - n\vec{a}}{m - n}$$

product where the result is a scalar, and vector (or cross) product where the result is a vector. Based upon these two types of products for vectors, they have found various applications in geometry, mechanics and engineering. In this section, we will discuss these two types of products.

10.6.1 Scalar (or dot) product of two vectors

Definition 2 The scalar product of two nonzero vectors $\vec{a}$ and $\vec{b}$, denoted by $\vec{a} \cdot \vec{b}$, is

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defined as

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta,$$

where, θ is the angle between $\vec{a}$ and $\vec{b}$, $0 \leq \theta \leq \pi$ (Fig 10.19).

If either $\vec{a} = 0$ or $\vec{b} = 0$ then θ is not defined, and in this case, we

define $\vec{a} \cdot \vec{b} = 0$

Observations

1. $\vec{a} \cdot \vec{b}$ is a real number.
2. Let $\vec{a}$ and $\vec{b}$ be two nonzero vectors, then $\vec{a} \cdot \vec{b} = 0$ if and only if $\vec{a}$ and $\vec{b}$ are perpendicular to each other. i.e.

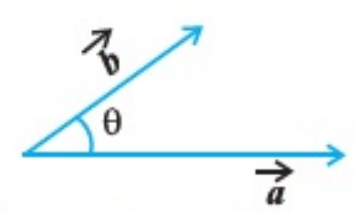
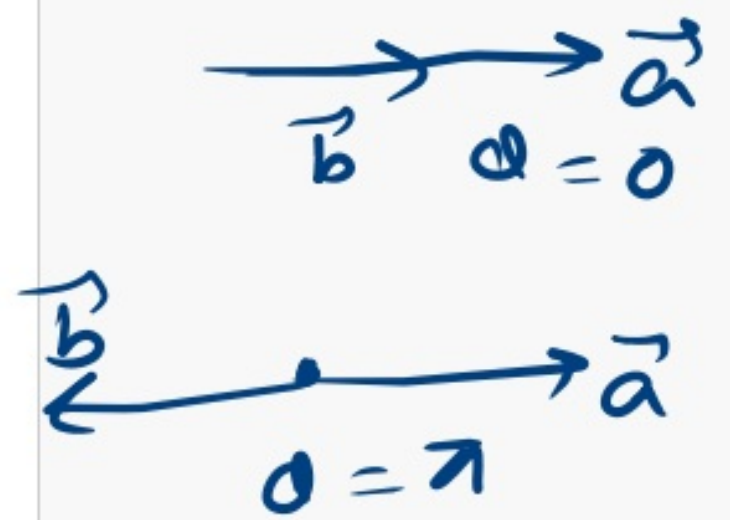


Fig 10.19

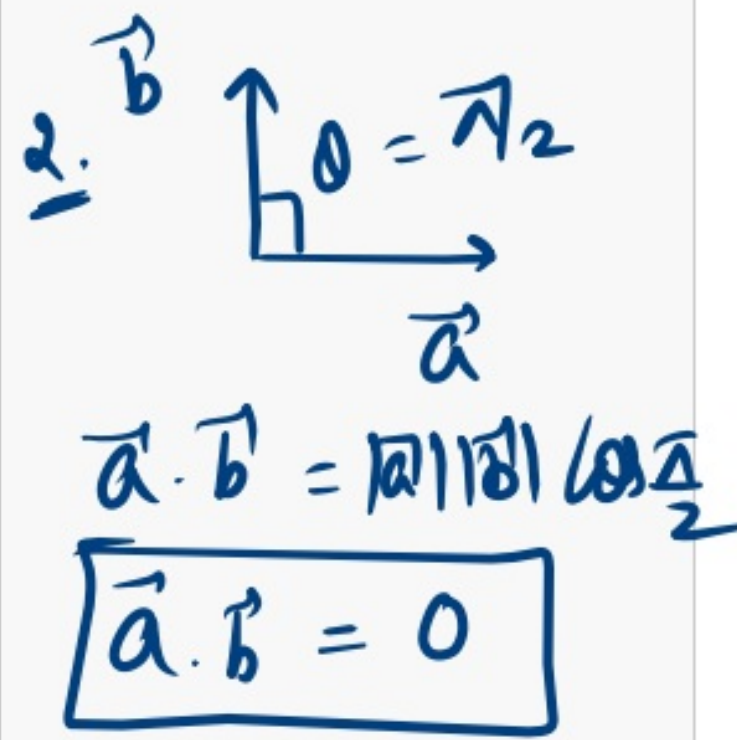


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define $\vec{a} \cdot \vec{b} = 0$

Observations/Note:-

1. $\vec{a} \cdot \vec{b}$ is a real number.
2. Let $\vec{a}$ and $\vec{b}$ be two nonzero vectors, then $\vec{a} \cdot \vec{b} = 0$ if and only if $\vec{a}$ and $\vec{b}$ are perpendicular to each other. i.e.

$$\boxed{\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}}$$

3. If $\theta = 0$, then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$
 In particular, $\vec{a} \cdot \vec{a} = |\vec{a}|^2$, as θ in this case is 0.

4. If $\theta = \pi$, then $\vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}|$
 In particular, $\vec{a} \cdot \vec{b} = -|\vec{a}||\vec{b}|$, as θ in this case is π .

5. In view of the Observations 2 and 3, for mutually perpendicular unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$, we have

$$\begin{aligned}\hat{i} \cdot \hat{i} &= \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1, \\ \hat{i} \cdot \hat{j} &= \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0\end{aligned}$$

6. The angle between two nonzero vectors $\vec{a}$ and $\vec{b}$ is given by

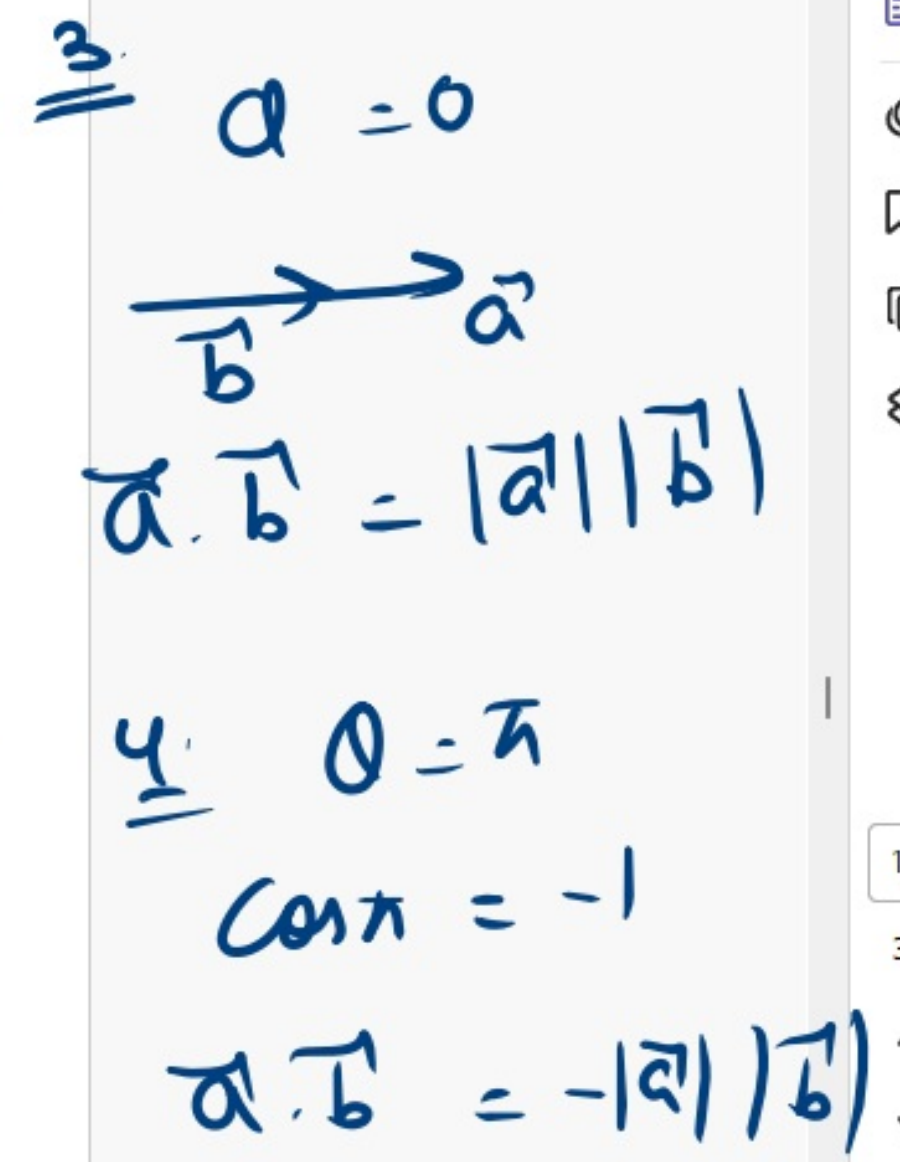
$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}, \text{ or } \theta = \cos^{-1}\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}\right)$$

7. The scalar product is commutative. i.e.

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} \quad (\text{Why?})$$

Two important properties of scalar product

Property 1 (Distributivity of scalar product over addition) Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be





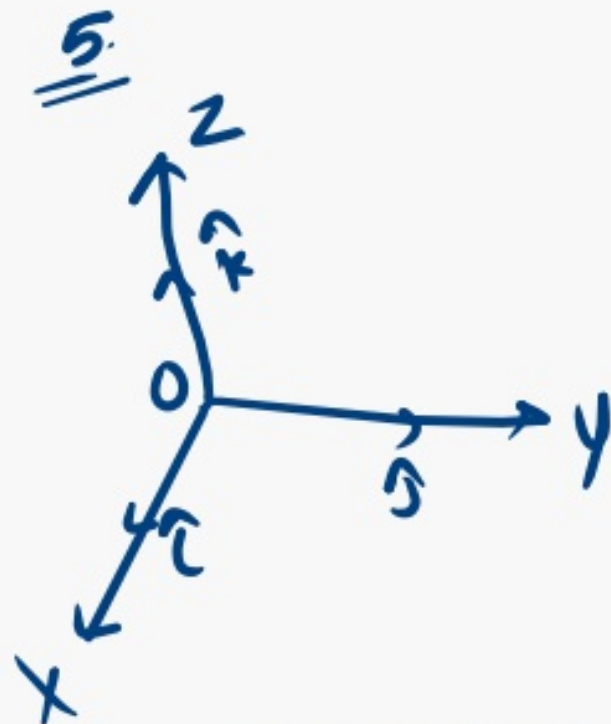
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$$\begin{aligned} \hat{i} \cdot \hat{j} &= \hat{j} \cdot \hat{k} = \hat{i} \cdot \hat{k} = 0 \\ \hat{i} \cdot \hat{i} &= \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1 \end{aligned}$$

define $\vec{a} \cdot \vec{b} = 0$

Observations

1. $\vec{a} \cdot \vec{b}$ is a real number.
2. Let $\vec{a}$ and $\vec{b}$ be two nonzero vectors, then $\vec{a} \cdot \vec{b} = 0$ if and only if $\vec{a}$ and $\vec{b}$ are perpendicular to each other. i.e.

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

3. If $\theta = 0$, then $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}|$

In particular, $\vec{a} \cdot \vec{a} = |\vec{a}|^2$, as θ in this case is 0.

4. If $\theta = \pi$, then $\vec{a} \cdot \vec{b} = -|\vec{a}| |\vec{b}|$

In particular, $\vec{a} \cdot \vec{b} = -|\vec{a}| |\vec{b}|$, as θ in this case is π .

5. In view of the Observations 2 and 3, for mutually perpendicular unit vectors

$\hat{i}$, $\hat{j}$ and $\hat{k}$, we have

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1,$$

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

6. The angle between two nonzero vectors $\vec{a}$ and $\vec{b}$ is given by

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}, \text{ or } \theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right)$$

7. The scalar product is commutative. i.e.

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a} \quad (\text{Why?})$$

Two important properties of scalar product

Property 1 (Distributivity of scalar product over addition) Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be

$$\begin{aligned} \hat{i} \cdot \hat{i} &= |\hat{i}| |\hat{i}| \cos 0 \\ &= 1 \cdot 1 \cdot 1 \end{aligned}$$

Free trial



7. $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$
 $\vec{b} \cdot \vec{a} = |\vec{b}| |\vec{a}| \cos \theta$
 scalar product
 is commutative.

$$\begin{aligned} \vec{a} \cdot (\vec{b} + \vec{c}) &= \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} \end{aligned}$$

$\hat{i}, \hat{j}$ and $\hat{k}$, we have
 $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1,$
 $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$

6. The angle between two nonzero vectors $\vec{a}$ and $\vec{b}$ is given by
 $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}, \text{ or } \theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right)$

7. The scalar product is commutative. i.e.
 $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$ (Why?)

Two important properties of scalar product
Property 1 (Distributivity of scalar product over addition) Let $\vec{a}, \vec{b}$ and $\vec{c}$ be any three vectors, then

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

$$\vec{a} \cdot (\lambda \vec{b}) = \lambda (\vec{a} \cdot \vec{b})$$

Property 2 Let $\vec{a}$ and $\vec{b}$ be any two vectors, and λ be any scalar. Then
 $(\lambda \vec{a}) \cdot \vec{b} = \lambda (\vec{a} \cdot \vec{b}) = \vec{a} \cdot (\lambda \vec{b})$

10.

$$\vec{a} = a_1 \hat{i} + b_1 \hat{j} + c_1 \hat{k}$$

$$\vec{b} = a_2 \hat{i} + b_2 \hat{j} + c_2 \hat{k}$$

$$[\vec{a} \cdot \vec{b}] = a_1 a_2 + b_1 b_2 + c_1 c_2$$

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$$\vec{a} \cdot \vec{a} = |\vec{a}| |\vec{a}| \cos 0$$
$$= |\vec{a}|^2$$



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eg 17

$$\vec{a} \cdot \vec{b} = (1)(1) + (1)(-1) + (-1)(1) = -1$$

$$|\vec{a}| = \sqrt{3}$$

$$|\vec{b}| = \sqrt{3}$$

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} = -\frac{1}{3} \Rightarrow \theta = \cos^{-1}\left(-\frac{1}{3}\right) \text{ AN}$$

$$\cos \alpha = \frac{\vec{a} \cdot \hat{i}}{|\vec{a}| |\hat{i}|} = \frac{a_1}{|\vec{a}|}, \cos \beta = \frac{a_2}{|\vec{a}|}, \text{ and } \cos \gamma = \frac{a_3}{|\vec{a}|}$$

Also, note that $|\vec{a}| \cos \alpha$, $|\vec{a}| \cos \beta$ and $|\vec{a}| \cos \gamma$ are respectively the projections of $\vec{a}$ along OX, OY and OZ. i.e., the scalar components a_1 , a_2 and a_3 of the vector $\vec{a}$, are precisely the projections of $\vec{a}$ along x-axis, y-axis and z-axis, respectively. Further, if $\vec{a}$ is a unit vector, then it may be expressed in terms of its direction cosines as

$$\vec{a} = \cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}$$

Example 13 Find the angle between two vectors $\vec{a}$ and $\vec{b}$ with magnitudes 1 and 2 respectively and when $\vec{a} \cdot \vec{b} = 1$.

Solution Given $\vec{a} \cdot \vec{b} = 1$, $|\vec{a}| = 1$ and $|\vec{b}| = 2$. We have

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|} \right) = \cos^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{3}$$

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Example 14 Find angle ' θ ' between the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

Solution The angle θ between two vectors $\vec{a}$ and $\vec{b}$ is given by

Free trial

Q8

$$\vec{a} \cdot \vec{b} = \frac{1}{2}$$

$$\theta = 60$$

$$\cos 60 = \frac{1}{2}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\frac{1}{2} = k \cdot k \cdot \frac{1}{2}$$

$$\frac{1}{2} = \frac{1}{2} k^2$$

$$k^2 = 1$$

$$k = 1$$

1. Find the angle between two vectors $\vec{a}$ and $\vec{b}$ with magnitudes $\sqrt{3}$ and 2, respectively having $\vec{a} \cdot \vec{b} = \sqrt{6}$.

2. Find the angle between the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$

3. Find the projection of the vector $\hat{i} - \hat{j}$ on the vector $\hat{i} + \hat{j}$.

4. Find the projection of the vector $\hat{i} + 3\hat{j} + 7\hat{k}$ on the vector $7\hat{i} - \hat{j} + 8\hat{k}$.

5. Show that each of the given three vectors is a unit vector:

$$\frac{1}{7}(2\hat{i} + 3\hat{j} + 6\hat{k}), \frac{1}{7}(3\hat{i} - 6\hat{j} + 2\hat{k}), \frac{1}{7}(6\hat{i} + 2\hat{j} - 3\hat{k})$$

Also, show that they are mutually perpendicular to each other.

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$$|\vec{a}| = |\vec{b}| = 1$$

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6. Find $|\vec{a}|$ and $|\vec{b}|$, if $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$ and $|\vec{a}| = 8|\vec{b}|$.

7. Evaluate the product $(3\vec{a} - 5\vec{b}) \cdot (2\vec{a} + 7\vec{b})$.

8. Find the magnitude of two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude and such that the angle between them is 60° and their scalar product is $\frac{1}{2}$.

9. Find $|\vec{x}|$, if for a unit vector $\vec{a}$, $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 12$.

Free trial

Q5

$$|\vec{a}| = \sqrt{\left(\frac{2}{7}\right)^2 + \left(\frac{13}{7}\right)^2 + \left(\frac{6}{7}\right)^2}$$

$$= \sqrt{\frac{4 + 169 + 36}{7^2}}$$

$$= \sqrt{\frac{189}{49}} = 1$$

$\vec{a} \rightarrow$ unit vector
 $\vec{b}$
 $\vec{c}$

- Find the angle between two vectors $\vec{a}$ and $\vec{b}$ with magnitudes $\sqrt{3}$ and 2, respectively having $\vec{a} \cdot \vec{b} = \sqrt{6}$.
- Find the angle between the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$
- Find the projection of the vector $\hat{i} - \hat{j}$ on the vector $\hat{i} + \hat{j}$.
- Find the projection of the vector $\hat{i} + 3\hat{j} + 7\hat{k}$ on the vector $7\hat{i} - \hat{j} + 8\hat{k}$.
- Show that each of the given three vectors is a unit vector:
 $\frac{1}{7}(2\hat{i} + 3\hat{j} + 6\hat{k})$, $\frac{1}{7}(3\hat{i} - 6\hat{j} + 2\hat{k})$, $\frac{1}{7}(6\hat{i} + 2\hat{j} - 3\hat{k})$
 Also, show that they are mutually perpendicular to each other.

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- Find $|\vec{a}|$ and $|\vec{b}|$, if $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$ and $|\vec{a}| = 8|\vec{b}|$.
- Evaluate the product $(3\vec{a} - 5\vec{b}) \cdot (2\vec{a} + 7\vec{b})$.
- Find the magnitude of two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude and such that the angle between them is 60° and their scalar product is $\frac{1}{2}$.
- Find $|\vec{x}|$, if for a unit vector $\vec{a}$, $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 12$.

$$\vec{a} \cdot \vec{b} = \left(\frac{1}{7}\right)\left(\frac{1}{7}\right)(6 + (-18) + 12) = 0$$

$$\Rightarrow \vec{a} \perp \vec{b}$$

$$\Rightarrow \vec{b} \cdot \vec{c}$$

$$\Rightarrow \vec{a} \cdot \vec{c}$$



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$$\vec{a} + \vec{b} = 6\hat{i} + 2\hat{j} - 8\hat{k}$$

$$\vec{a} - \vec{b} = 4\hat{i} - 4\hat{j} + 2\hat{k}$$

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 24 + (-8) + (-16) = 0$$

$$\Rightarrow (\vec{a} + \vec{b}) \perp (\vec{a} - \vec{b})$$

Free trial

Example 14 Find angle ' θ ' between the vectors $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

Solution The angle θ between two vectors $\vec{a}$ and $\vec{b}$ is given by

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

$$\vec{a} \cdot \vec{b} = (\hat{i} + \hat{j} - \hat{k}) \cdot (\hat{i} - \hat{j} + \hat{k}) = 1 - 1 - 1 = -1.$$

$$\cos \theta = \frac{-1}{3}$$

Therefore, we have

$$\text{hence the required angle is } \theta = \cos^{-1} \left(-\frac{1}{3} \right)$$

Example 15 If $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, then show that the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular.

Solution We know that two nonzero vectors are perpendicular if their scalar product is zero.

$$\text{Here } \vec{a} + \vec{b} = (5\hat{i} - \hat{j} - 3\hat{k}) + (\hat{i} + 3\hat{j} - 5\hat{k}) = 6\hat{i} + 2\hat{j} - 8\hat{k}$$

$$\text{and } \vec{a} - \vec{b} = (5\hat{i} - \hat{j} - 3\hat{k}) - (\hat{i} + 3\hat{j} - 5\hat{k}) = 4\hat{i} - 4\hat{j} + 2\hat{k}$$

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = (6\hat{i} + 2\hat{j} - 8\hat{k}) \cdot (4\hat{i} - 4\hat{j} + 2\hat{k}) = 24 - 8 - 16 = 0.$$

Hence $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular vectors.

Example 16 Find the projection of the vector $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on the vector $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$.

Solution The projection of vector $\vec{a}$ on the vector $\vec{b}$ is given by



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Q10

$$\vec{a} + \lambda \vec{b} = 2\hat{i} + 2\hat{j} + 3\hat{k} + \lambda(-\hat{i} + 2\hat{j} + \hat{k})$$

$$= \hat{i}(2 - \lambda) + \hat{j}(2 + 2\lambda) + \hat{k}(3 + \lambda)$$

$$(\vec{a} + \lambda \vec{b}) \perp \vec{c}$$

Free trial

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6. Find $|\vec{a}|$ and $|\vec{b}|$, if $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 8$ and $|\vec{a}| = 8|\vec{b}|$.
7. Evaluate the product $(3\vec{a} - 5\vec{b}) \cdot (2\vec{a} + 7\vec{b})$.
8. Find the magnitude of two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude and such that the angle between them is 60° and their scalar product is $\frac{1}{2}$.
9. Find $|\vec{x}|$, if for a unit vector $\vec{a}$, $(\vec{x} - \vec{a}) \cdot (\vec{x} + \vec{a}) = 12$.
10. If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$ are such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$, then find the value of λ .
11. Show that $|\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ is perpendicular to $|\vec{a}|\vec{b} - |\vec{b}|\vec{a}$, for any two nonzero vectors $\vec{a}$ and $\vec{b}$.
12. If $\vec{a} \cdot \vec{i} = 0$ and $\vec{a} \cdot \vec{b} = 0$, then what can be concluded about the vector $\vec{b}$?
13. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, find the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.
14. If either vector $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then $\vec{a} \cdot \vec{b} = 0$. But the converse need not be true. Justify your answer with an example.
15. If the vertices A, B, C of a triangle ABC are (1, 2, 3), (-1, 0, 0), (0, 1, 2), respectively, then find $\angle ABC$. [$\angle ABC$ is the angle between the vectors $\vec{BA}$ and $\vec{BC}$].
16. Show that the points A(1, 2, 7), B(2, 6, 3) and C(3, 10, -1) are collinear.

$$\Rightarrow (\vec{a} + \lambda \vec{b}) \cdot \vec{c} = 0$$

$$3(2 - \lambda) + (1)(2 + 2\lambda) + (0)(3 + \lambda) = 0$$

$$6 - 3\lambda + 2 + 2\lambda = 0$$

$$8 - \lambda = 0$$

$$\boxed{8 = \lambda}$$

perpendicular to $\vec{c}$, then find the value of n .

11. Show that $|\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ is perpendicular to $|\vec{a}|\vec{b} - |\vec{b}|\vec{a}$, for any two nonzero vectors $\vec{a}$ and $\vec{b}$.

Property

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

$$\text{say, } \vec{c} = |\vec{a}|\vec{b} + |\vec{b}|\vec{a}$$

$$\text{say, } \vec{d} = |\vec{a}|\vec{b} - |\vec{b}|\vec{a}$$

$$\vec{c} \cdot \vec{d} = (|\vec{a}|\vec{b} + |\vec{b}|\vec{a}) \cdot (|\vec{a}|\vec{b} - |\vec{b}|\vec{a})$$

$$\vec{c} \cdot \vec{d} = |\vec{a}|\vec{b} \cdot (|\vec{a}|\vec{b} - |\vec{b}|\vec{a}) + |\vec{b}|\vec{a} \cdot (|\vec{a}|\vec{b} - |\vec{b}|\vec{a})$$

$$\vec{c} \cdot \vec{d} = |\vec{a}|^2 \vec{b} \cdot \vec{b} - |\vec{a}||\vec{b}|\vec{b} \cdot \vec{a} + |\vec{b}||\vec{a}|\vec{a} \cdot \vec{b} - |\vec{b}|^2 \vec{a} \cdot \vec{a}$$

$$= \cancel{|\vec{a}|^2 |\vec{b}|^2} - \cancel{|\vec{a}||\vec{b}|\vec{a} \cdot \vec{b}} + \cancel{|\vec{b}||\vec{a}|\vec{a} \cdot \vec{b}} - \cancel{|\vec{b}|^2 |\vec{a}|^2}$$

$$= 0$$

$$\Rightarrow \vec{c} \perp \vec{d}$$

Homework

Ex 10.2

Q8, Q5, Q11, Q12, Q13, Q14, Q15, Q16, Q17, Q18, Q19

Example - 11, 12, 15, 18

Ex 10.3 — Q1, Q2, Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q18

+
Exercise 7.7 (all questions)